COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 1 : Session – 4 : SLO – 2 | Codes and parity
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5 MCQs

• 1. b — 1110 (Gray of 1011: 1, 1^0=1, 0^1=1, 1^1=0)
• 2. c — 1010 (7 + 3 = 10)
• 3. a — 0000 (0) with carry (9+3=12 → 1001+0011=1100, +0110 → 0010 carry 1... i.e., BCD 12; as single digit it is 0010 with carry)
• 4. a — 65
• 5. b — Ensures total number of 1s is even

2 Two-Mark Questions

• 1. Gray 10011 to binary: b4 = g4 = 1; b3 = b4^g3 = 1^0 = 1; b2 = b3^g2 = 1^0 = 1; b1 = b2^g1 = 1^1 = 0; b0 = b1^g0 = 0^1 = 1. Binary = 11011.
• 2. Decimal 2: BCD = 0010; Excess-3 = 0010 + 0011 = 0101.

1 Scenario-Based Long Answer

• Pipeline: take decimal digit 5 → BCD 0101 → Excess-3 1000 → count 1s in 1000 = 1 (odd) → even-parity bit = 1 → transmitted frame 1 1000.
• At the receiver, the 1s in the 5-bit frame are counted: 11000 has two 1s (even) → no error. If a bit flips in transit (e.g. 11010, three 1s), the parity check fails and the digit is rejected/retransmitted. BCD→XS-3→parity thus gives both decimal-friendly encoding and single-bit error protection.

5 Match the Following

• Binary to Gray — a) Copy first bit, then XOR
• ASCII of 'B' — b) 66
• BCD Addition — c) Add 0110 if result > 1001
• XS-3 code for 2 — d) 0101
• Even Parity — e) Total number of 1s is even

5 Fill in the Blanks

• 1. 10101 (Gray of 11001: 1, 1^1=0, 1^0=1, 0^0=0, 0^1=1)
• 2. B 3. 0110 4. 0110 (3+3=6) 5. error

1 Scenario-Based Small Activity

• Data 1100110: count of 1s = 4 (already even) → even parity bit = 0.
• Appended as MSB, final 8-bit result: 01100110.

5 True or False

• 1. FALSE — Standard ASCII uses 7 bits. 2. TRUE 3. TRUE
• 4. FALSE — Excess-3 is non-weighted. 5. FALSE — Even parity ensures an even number of 1s.